Relationships and Foreign Keys

Relational Data Model

Product

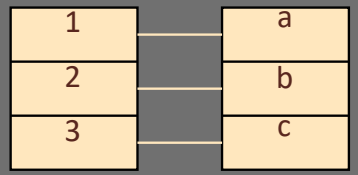
prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

Multiple relations.
No pointers.
Connections via repeated data.

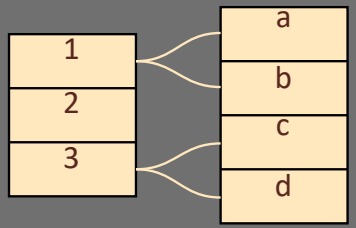
Manufacturer

man_name	man_address	man_city	man_state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

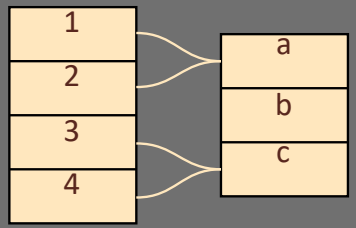
Types of relationships



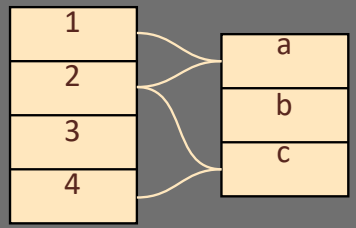
One-to-one



One-to-many



Many-to-one



Many-to-many

Foreign key

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

Attribute set that refers to a key in another table.

Manufacturer

man_name	man_address	man_city	man_state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

Referential Integrity

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

Enforce:

- Each FK value must be one of the key values.
- Each prod_manufacturer must be a man_name.

Manufacturer

man_name	man_address	man_city	man_state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

Referential Integrity & NULL

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

Enforce:

- Each FK value must be one of the key values.
- But that still allows FK to be NULL!

Manufacturer

man_name	man_address	man_city	man_state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

FKs vs. pointers

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

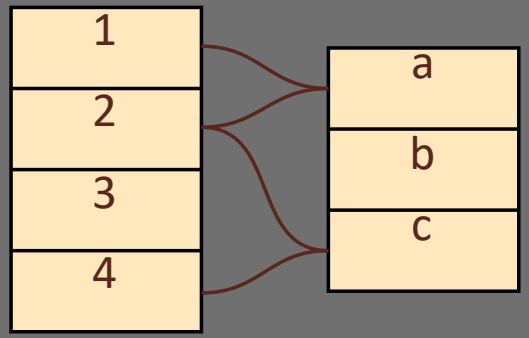
FK repeats data. Pointer points at location.

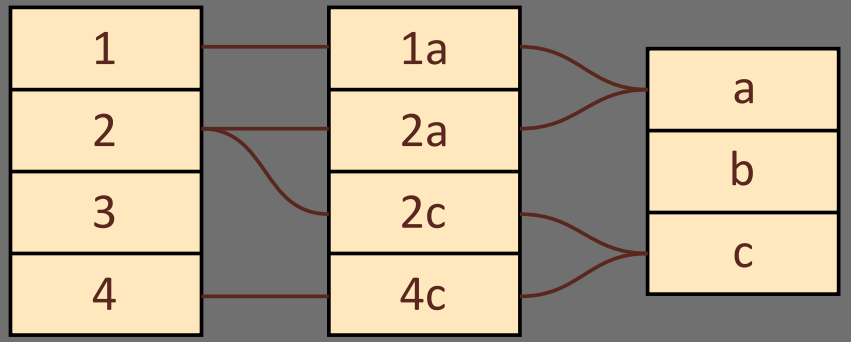
- Pointers are difficult to maintain when data changes.
- Pointers are uni-directional.

Manufacturer

man_name	man_address	man_city	man_state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

Encoding many-many with one-many





Using junction table

Example junction table

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

Sells

prod_name	st_id
Gizmo	1
Gizmo	3
PowerGizmo	2
Widget	1
Widget	2

Store

st_id	st_city
1	Houston
2	Austin
3	Dallas

Each Product is sold in many Stores.

Each Store sells many Products.

Database schema

Product(prod_name, prod_price, prod_manufacturer)

Manufacturer(man_name, man_address, man_city, man_state)